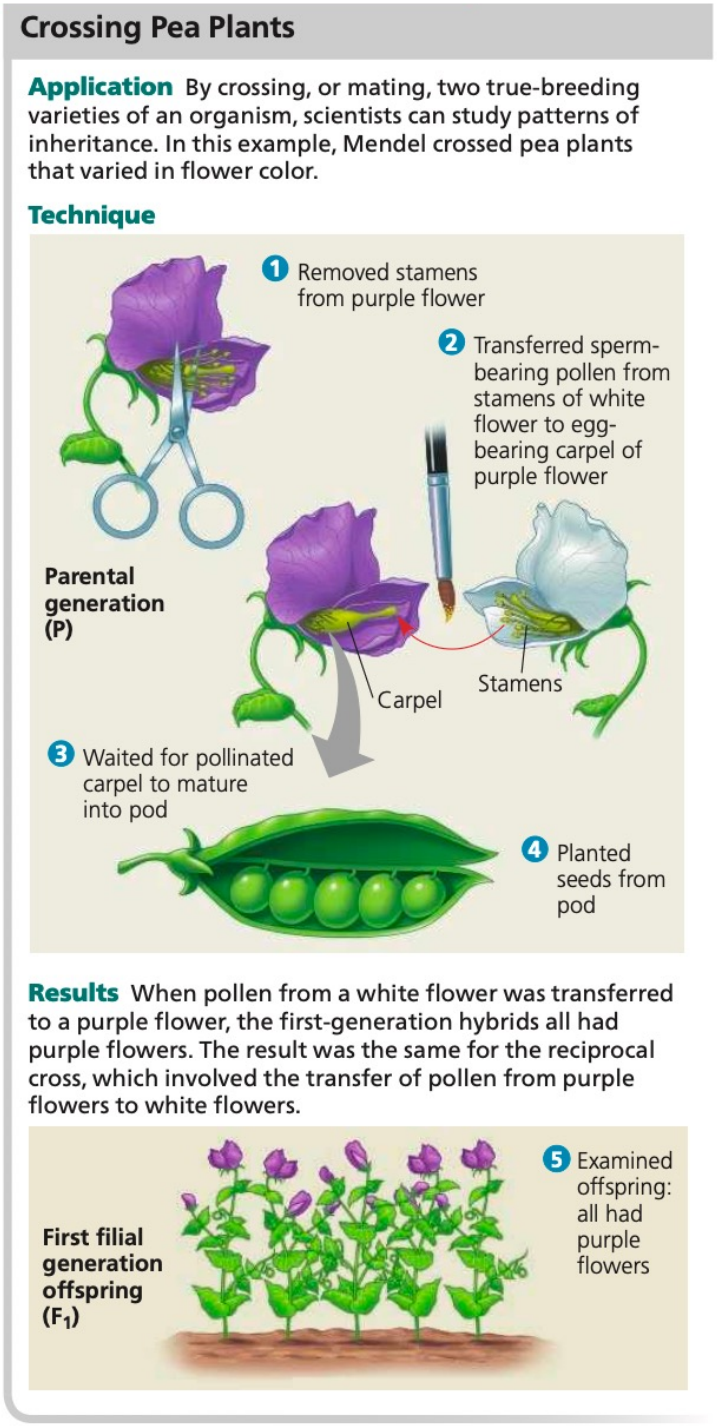
Project
PRISM

Mendelian Inheritance

Genetics and Evolution

What is Mendelian Inheritance?

- Mendelian inheritance refers to the patterns of inheritance first described by **Gregor Mendel** through experiments on pea plants.
- Mendel proposed that traits are controlled by discrete heritable units, now known as **genes**.
- Different versions of the same gene are called **alleles**.
- Mendelian inheritance explains how alleles are passed from parents to offspring according to predictable rules.
- Mendel's work established two main laws:
- the **law of segregation**
- the **law of independent assortment**.



Why Did Mendel Use Pea Plants?

- Mendel chose **garden peas** because they were easy to grow and had many clearly distinguishable traits.
- Pea plants normally **self-fertilize**, allowing Mendel to maintain **true-breeding** lines.
- He could also control fertilization manually to perform specific crosses.
- The traits he selected had relatively simple inheritance patterns, making them suitable for quantitative analysis.
- His careful counting of large numbers of offspring allowed him to detect consistent inheritance ratios.

Dominant and Recessive Alleles

- In Mendel's pea plants, one allele could mask the effect of another.
- The allele that determines the phenotype in a heterozygote is called **dominant**.
- The allele whose effect is hidden in a heterozygote is called **recessive**.
- For example, in pea flower colour:

$$\mathbf{P} = \text{purple (dominant)}$$
$$\mathbf{p} = \text{white (recessive)}$$

Thus:

- **PP** and **Pp** both have purple flowers
- **pp** has white flowers.

Mendel's Monohybrid Cross

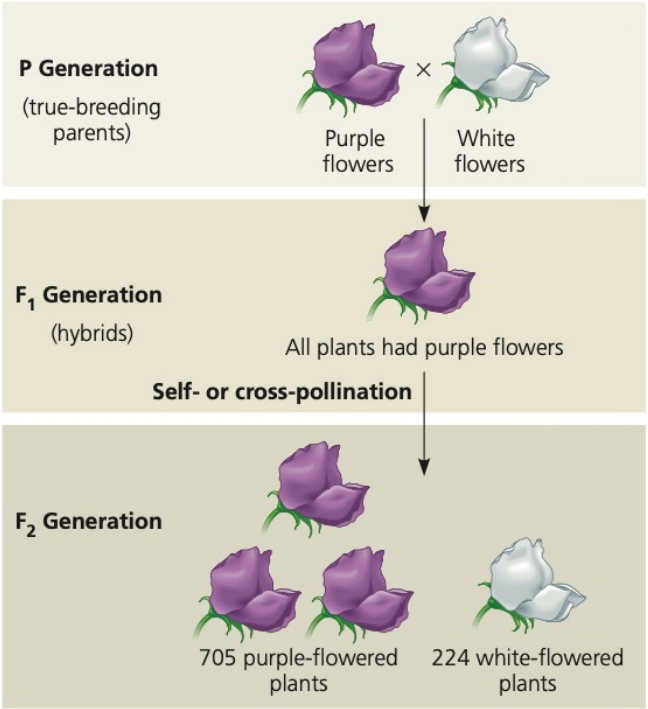
- A **monohybrid cross** follows the inheritance of **one character.**
- Mendel crossed two true-breeding parents with contrasting forms of one trait.
- Example: **PP × pp**
- All **F1 offspring** were **Pp** and showed the dominant phenotype.
- When the F1 generation self-fertilized, the **F2 generation** showed:

  3:1 phenotypic ratio

  1:2:1 genotypic ratio
- This revealed that inheritance is particulate; not blending.

When F_1 hybrid pea plants self- or cross-pollinate, which traits appear in the F_2 generation?

Experiment Mendel crossed true-breeding purple-flowered plants and white-flowered plants (crosses are symbolized by ×). The resulting F_1 hybrids were allowed to self-pollinate or were cross-pollinated with other F_1 hybrids. The F_2 generation plants were then observed for flower color.



Results Both purple-flowered and white-flowered plants appeared in the F_2 generation, in a ratio of approximately 3:1.

Conclusion The "heritable factor" for the recessive trait (white flowers) had not been destroyed, deleted, or "blended" in the F_1 generation but was merely masked by the presence of the factor for purple flowers, which is the dominant trait.

Data from G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn* 4:3–47 (1866).

WHAT IF? *If you mated two purple-flowered plants from the P generation, what ratio of traits would you expect to observe in the offspring? Explain. What might Mendel have concluded if he stopped his experiment after the F_1 generation?*

The Law of Segregation

- The **law of segregation** states that the two alleles for a gene separate during **gamete formation**.
- Each gamete receives only **one allele** from each gene pair.
- At fertilization, offspring receive one allele from each parent, restoring the pair.
- This explains why recessive traits can disappear in one generation and reappear in the next.
- The law of segregation is the basis for the **3:1 ratio** seen in Mendel's monohybrid crosses.

The Testcross

- An organism with a dominant phenotype may be either:
  homozygous dominant, or
  heterozygous
- To determine its genotype, a **testcross** is performed.
- In a testcross, the unknown individual is crossed with a **homozygous recessive** individual.

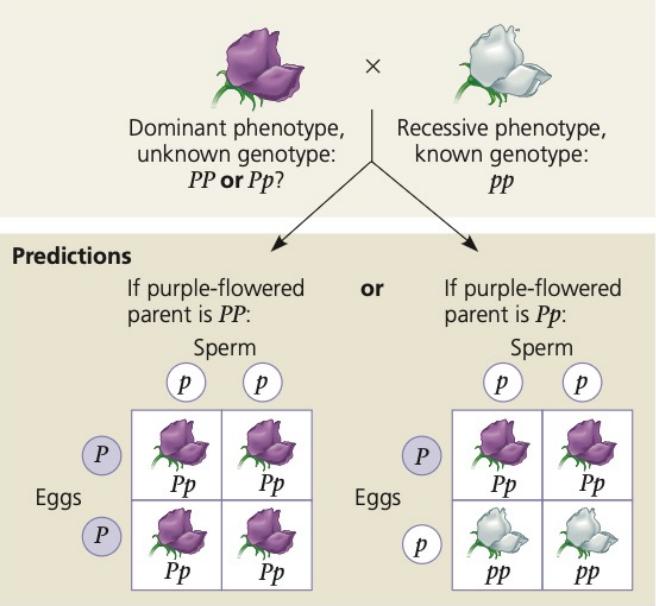
Example:

- If the unknown purple plant is **PP**, all offspring will be purple.
- If it is **Pp**, the offspring will show a **1:1 ratio** of purple to white.
- Therefore, a testcross reveals the genotype of an individual showing the dominant phenotype.

The Testcross

Application An organism that shows a dominant trait in its phenotype, such as purple flowers in pea plants, can be either homozygous for the dominant allele or heterozygous. To determine the organism's genotype, geneticists can perform a testcross.

Technique In a testcross, the individual with the unknown genotype is crossed with a homozygous individual expressing the recessive trait (white flowers in this example), and Punnett squares are used to predict the possible outcomes.



Results Matching the results to either prediction identifies the unknown parental genotype (either *PP* or *Pp* in this example). In this testcross, we transferred pollen from a white-flowered plant to the carpels of a purple-flowered plant; the opposite (reciprocal) cross would have led to the same results.

Mastering Biology **Animation: Testcross in "MendAliens"**

Dihybrid Crosses

- A **dihybrid cross** follows the inheritance of **two characters at the same time**.
- Example:

$$\mathbf{YYRR \times yyrr}$$

- All F1 offspring are **YyRr**, showing both dominant traits.
- When two F1 dihybrids are crossed, the F2 phenotypic ratio is:

$$\mathbf{9 : 3 : 3 : 1}$$

- This result led Mendel to formulate his second law of inheritance.

The Law of Independent Assortm

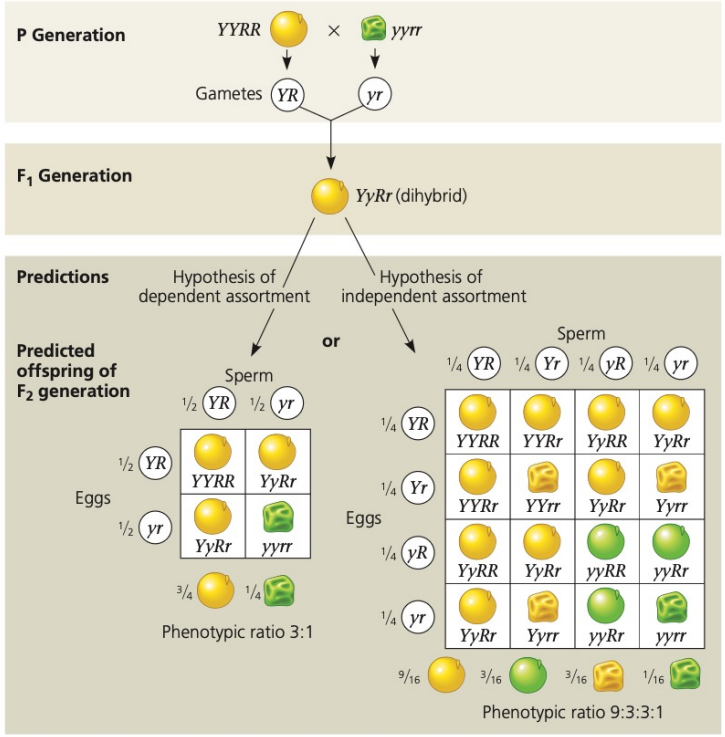
- The **law of independent assortment** states that alleles of one gene segregate independently of alleles of another gene during gamete formation.
- Thus, inheritance of one character usually does not affect inheritance of another.
- In a dihybrid individual such as **YyRr**, four types of gametes can be produced:

YR, Yr, yR, yr

- If assortment is independent, these gametes are formed in equal proportions.
- This explains the **9:3:3:1** phenotypic ratio in Mendel's dihybrid crosses.

Do the alleles for one character segregate into gametes dependently or independently of the alleles for a different character?

Experiment To follow the characters of seed color and seed shape through the F_2 generation, Mendel crossed a true-breeding plant with yellow round seeds with a true-breeding plant with green wrinkled seeds, producing dihybrid F_1 plants. Self-pollination of the F_1 dihybrids produced the F_2 generation. The two hypotheses (dependent and independent "assortment" of the two genes) predict different phenotypic ratios.



Results

315 108 101 32 Phenotypic ratio approximately 9:3:3:1

Conclusion The results support the hypothesis of independent assortment, the only one that predicts two newly observed phenotypes: green round seeds and yellow wrinkled seeds (see the right-hand Punnett square). The alleles for each gene segregate independently of those of the other, and the two genes are said to assort independently.

Data from G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn* 4:3–47 (1866).

WHAT IF? *Suppose Mendel had transferred pollen from an F_1 plant to the carpel of a plant that was homozygous recessive for both genes. Set up the cross and draw Punnett squares that predict the offspring for both hypotheses. Would this cross have supported the hypothesis of independent assortment equally well?*

Mastering Biology
Animation: Mendel's Cross of Two Characters: Seed Shape and Seed Color
Animation: A Simplified Cross of Two Characters in Humans
Animation: Crosses of Two Characters in "MendAliens"

Mendelian Inheritance and Probability

- Mendelian inheritance follows the **rules of probability**.
- The **multiplication rule** is used when two or more independent events occur together.
- Example:
- Probability of **YY** from $\mathbf{Yy} \times \mathbf{Yy} = \frac{1}{4}$
- Probability of **RR** from $\mathbf{Rr} \times \mathbf{Rr} = \frac{1}{4}$
- Probability of $\mathbf{YYRR} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$
- Probability can often be used instead of drawing very large Punnett squares.

Limits of Mendelian Inheritance

Mendelian inheritance works best for traits controlled by **one gene with two alleles**, with one allele completely dominant over the other.

This makes it useful as a foundational model.

However, not all traits follow simple Mendelian patterns.

More complex cases include:

- incomplete dominance
- codominance
- multiple alleles
- gene interaction
- sex linkage
- linkage between genes

Conclusion

- Mendelian inheritance explains how alleles are transmitted from parents to offspring in predictable patterns.
- Mendel showed that inheritance is based on **discrete units**, not blending.
- His work established:
- the **law of segregation**
- the **law of independent assortment**
- Monohybrid and dihybrid crosses provide the classic Mendelian ratios:

  3:1

  1:2:1

  9:3:3:1
- Mendelian inheritance forms the basis for later topics in genetics.

IBO past year questions

IBO 2013 Theoretical Exams 2: Qn 30.

In a plant species, the level of anthocyanin pigments produced is controlled by a single gene G, for which only a "dark" and a "light" allele are present. To more accurately map the position of gene G on chromosome 3, two inbred lines (P1 and P2) are crossed and F2 individuals (X1 through X5) are genotyped at five single nucleotide variant loci (SNV1 through SNV5) on the same chromosome.

	SNV1	SNV2	SNV3	SNV4	SNV5	Anthocyanin [mM]
P1	A/A	C/C	A/A	T/T	C/C	120
P2	T/T	T/T	C/C	G/G	G/G	25
X1	A/T	C/T	A/C	T/G	C/G	115
X2	A/A	C/C	A/A	T/G	C/G	123
X3	A/A	T/T	C/C	T/G	C/G	22
X4	A/A	C/T	C/C	T/G	C/C	29
X5	A/T	C/T	A/A	T/T	C/C	118

Based on these results, indicate if each of the following statements is true or false.

A. One recombination event happened in each parent of X2 between the genotyped loci.

B. F1 individuals are likely showing intermediate levels of anthocyanin.

C. Among the studied loci, SNV3 is closest to gene G.

D. The phenotype ratio in the progeny of a cross between X4 and X5 is 2:1.

IBO 2016 Theoretical Exam B Qn 84

An organism has four genes, A, B, C and D with two alleles each. An individual heterozygous for these genes was bred with one that is homozygous recessive. The cross produced 3288 offspring with phenotypes shown in the table below:

Phenotypes	Number of individuals
ABCD	675
ABCd	83
ABcD	1
ABcd	74
AbCD	73
AbCd	1
AbcD	84
Abcd	670
aBCD	655
aBCd	86
aBcD	1
aBcd	73
abCD	71
abCd	1
abcD	87
abcd	653

Indicate in the answer sheet if each of the following statements is true or false.

A. The four loci are genetically linked.

B. The distance between gene B and gene D is 9 cM.

C. The distance between gene D and gene C is 10.5 cM.

D. Interference happened with a value less than 0.25. Interference = 1-(observed frequency of double crossover/expected frequency of double crossover).

Qn 85

A plant is normally red-flowered. Plant breeders obtained three genetically different pure mutated lines of white-flowered plants (designated as a, b and c). They performed crosses and observed progeny phenotypes as follows:

Cross	Parent	Progeny
1	line a x line b	F_1 all white
2	line a x line c	F_1 all red
3	line b x line c	F_1 all white
4	red F_1 from cross 2 x line a	1/4 red : 3/4 white
5	red F_1 from cross 2 x line b	1/8 red : 7/8 white
6	red F_1 from cross 2 x line c	1/2 red : 1/2 white

Indicate in the answer sheet if each of the following statements is true or false.

A. Line (a) has only one homozygous mutated gene.

B. Line (b) shares two homozygous mutated genes with line c.

C. Line (c) shares one homozygous mutated gene with line a.

D. Line (b) has three homozygous mutated genes.

Qn 90

Pedigrees 1-4 show the inheritance of four different **rare** disorders. It is known that the disease in pedigree 4 is X-linked recessive.

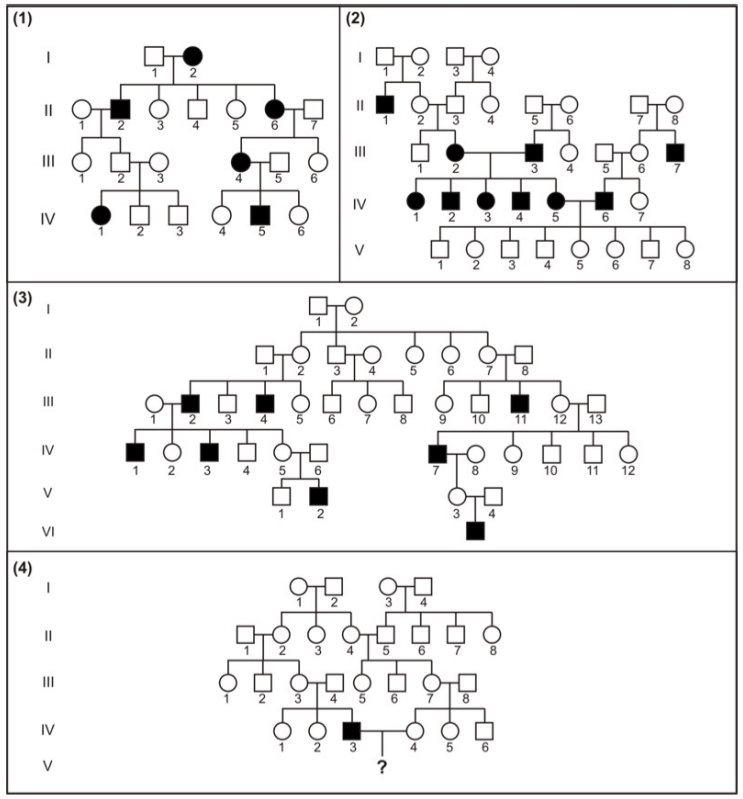


Figure Q.90

Studying the pedigrees and indicate in the answer sheet if each of the following statements is true or false.

A. The disorder in pedigree 1 is most likely caused by a recessive allele.

B. Person III_2 and III_7 in pedigree 2 have the same genotype.

C. Pedigree 3 shows the inheritance of the disorder can be caused by a recessive allele on X-chromosome.

D. If the affected man and his unaffected wife in pedigree 4 have a son then the probability of this son be affected is 0.125.

Qn 91

Polymorphic DNA sequences are widely used for molecular identification. Short tandem repeat (STR) is composed of multiple repeats of 2-8 nucleotides flanked by two conserved sequences. Each STR locus normally has more than two alleles. Single nucleotide polymorphism (SNP) is a variation at a single position in a DNA sequence among individuals. Each SNP usually has only two alleles. Seven individuals were genotyped for two autosomal and two mitochondrial (mtDNA) SNPs, two autosomal and two Y-linked (NRY) STRs (**Table Q.91**).

	Autosomes				NRY		mtDNA	
Individuals	SNP1	SNP2	STR1	STR2	STR1	STR2	SNP1	SNP2
Ind_1	A/A	A/A	13/15	18/20	13	12	C	A
Ind_2	A/C	A/G	12/14	18/21	13	15	T	A
Ind_3	C/C	A/G	14/15	18/21	13	15	C	G
Ind_4	A/C	G/G	13/15	19/19	11	14	T	G
Ind_5	C/C	A/G	14/15	18/19	-	-	C	G
Ind_6	A/C	G/G	14/14	18/19	-	-	T	G
Ind_7	C/C	G/G	14/16	19/21	-	-	C	A

Table Q.91

Indicate in the answer sheet if each of the following statements is true or false.

A. If the same number of SNPs or STRs are used, SNPs are better than STRs for distinguishing individuals.

B. Ind_6 is more likely a child of Ind_2 and Ind_5 than Ind_3 is.

C. Ind_4 is possibly a brother of Ind_6.

D. It is possible that Ind_7 is a granddaughter of Ind_1 and Ind_6.

Answer Key

IBO Theoretical Exam 2: Qn 30

A. False **B.** False **C.** True **D.** False

Original commentary
Correct answers
A *false*
Most likely, a single recombination event happened in only one of the parents, as one of the haplotype is parental (ACATC) and the other shows a recombination between SNV3 and SNV4.
B *false*
Gene G is linked to SNV3, and the table lists a heterozygous individual (X1) with an elevated anthocyanin level (which is the dominant allele). But note that the students do not need to identify the most closely linked locus as for each of them heterozygous individuals are given and the conclusion would remain unchanged.
C *true*
This is the only locus for which the genotypes match the phenotypes in a Mendelian fashion.
D *false*
A 2:1 ratio is impossible for any crossing, as we clearly have dominant-recessive inheritance. So it is possible to answer this question even if the truly linked locus is not identified.

IBO Theoretical Exam B: Qn 84

A. False B. False C. True D. False

Explanation

A. False. Genes BCD are linked but gene A is unlinked with them.

B. False. The distance B-D: 100% (83+74+73+84+86+73+71+87)/3288 = 19 m.u.

C. True. The distance C-D: 100% (87+1+1+86+84+1+1+83)/3288 = 10.5 m.u.

D. False. Interference (I) = 1- [(observed DCO)/(expected DCO)] = 1- {[(1+1+1+1)/3288]/[(0.0897) (0.105)]} = 0.89.

Qn 85

Answer key

A. False B. False C. False D. True

Explanation

Note: To solve this problem, based on complementation tests students should determine how many genes involved.

Cross 2 tells that there are at least two genes involved because white crossed white yields red flowers. However, cross 5 indicates that there are at least three genes involved because a 1: 7 ratio is observed. Therefore, F_1 red must be heterozygous for 3 genes and line b must be homozygous for three defective genes located in different chromosomes (**D is true**).

Compare crosses 2 and 3. Lines (a) and (c) can compensate for each other's defects but lines (b) and (c) cannot. This suggests that line (b) shares a defective homozygous gene with line (c) (**B is false**) and that line (a) is normal for the defective gene seen in line (c) but not in line (b). Line a must be homozygous for 2 defective genes (**A is false**). Line (a) and line (c) compensate for each other's defect so these lines cannot share any defective genes (**C is false**). Line (c) is normal for the two defective genes in line (a). Genotypes of the three lines should be: Line a: aaBBcc; Line b: aabbcc; line c: AAbbCC.

Qn 90

Answer key

A. False B. False C. True D. False

Explanation

A. False. If the disorder is autosomal recessive then three people I_1, II_7 and III_5 must be carriers. Heterozygotes should not be so common because it is a rare disorder. Therefore, it is not likely the disorder is caused by a recessive allele. The disorder is actually caused by a dominant allele.

B. False. The disorder is caused by recessive alleles. Persons IV_5 and IV_6 were recessive homozygotes but none of their children was affected. It indicates that there is complementation between recessive alleles of different genes. Therefore, the genotype of person III_2 must be different from the genotype of person III_7.

C. True.

D. False. Pedigree 5 shows that the disorder was caused by a recessive allele on X-chromosome. In the pedigree 4, the male IV_3 is affected therefore females III_3, II_2 and I_1 must be carriers. Chance of the recessive alleles be passed on from female I_1 to female II_4 is ½, from female II_4 to female III_7 is ½, from female III_7 to female IV_4 is ½ and from female IV_4 to her son is ½. Therefore we can calculate the probability for the son to be affected is ½ * ½ * ½ * ½ = 0.0625.

Qn 91

Answer key

A. False B. False C. True D. True

Explanation

A. False. SNPs are less polymorphic than STRs therefore they are not as good as STRs to differentiate individuals of the same species.

B. False. All autosomal markers of Ind_3 and Ind_6 can be inherited from Ind_2 and Ind_5. Ind_3 share Y chromosome markers with Ind_2 and mitochondrial markers with Ind_5 while Ind_6 does not share mitochondrial marker with Ind_5.

C. True. Ind_4 and Ind_6 share autosomal SNP markers and mitochondrial markers. Their autosomal STR1 and STR2 can be inherited from parents of 13/14 and 14/15, and 18/19 and 19/19, respectively.

D. True. It is not necessary that Ind_7 inherits mitochondrial marker from Ind_6. It is possible that Ind_7 inherits all autosomal markers from her father who is a son of Ind_1 and Ind_6.

End of slides